

Figure 4.2 The RQ-1 Predator (1995) – design and operational architecture.⏎

The labeled diagram highlights major features, including the composite airframe, pusher propeller configuration, sensor turret (EO/IR camera system) located under the nose, and synthetic aperture radar (SAR) mounted beneath the fuselage. The satellite communication (SATCOM) antenna positioned on the upper fuselage enables long-distance, beyond-line-of-sight (BLOS) control via ground stations. The Rotax 914 piston engine at the rear provides propulsion, while the fixed tricycle landing gear supports stable takeoff and landing operations.

This configuration allowed the Predator to conduct long-endurance surveillance missions with exceptional stability and precision. The EO/IR turret provided real-time video feeds for reconnaissance and target tracking, while the laser designator enabled precision-guided weapon deployment in the upgraded MQ-1 variant. The diagram also emphasizes the ground control station (GCS) link, demonstrating the closed-loop data transmission between the UAV and its remote operators through secure SATCOM channels.